Fiscal Year:	FY 2004	**Task Last Updated:**	FY 08/30/2004
PI Name:	Prosperetti, Andrea Ph.D.		
Project Title:	Acoustic Bubble Removal from Boiling Surfaces and its Optimization		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	03/07/2000	**End Date:**	11/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ohsaka, Kenichi (University of Southern California)		
Grant/Contract No.:	NAG3-2362		
Performance Goal No.:			
Performance Goal Text:	note - end date changed from 11/30/2003		
Task Description:	In a microgravity environment vapor bubbles tend to remain near the boiling surface where they are generated. This circumstance affects the boiling heat transfer and in particular promotes an early transition to the highly inefficient film boiling regime. The purpose of the present project is a combined theoretical and experimental study of the effectiveness of acoustic means in removing bubbles from the boiling surface by exploiting acoustic pressure-radiation (or Bjerknes) forces. The objective of the theoretical part of the work is the development of computational tools to simulate the behavior of vapor bubbles in acoustic and flow fields, with the purpose of using these tools to understand the physical principles at play and enhance heat transfer capabilities in space. The objective of the experimental part is to examine the behavior of vapor bubbles levitated in a sound field.		

	Boiling is one of the most effective heat transfer mechanisms available and, if it were possible to exploit and control it reliably in a microgravity environment, great savings in cost and weight would be realized. Unfortunately, Earth gravity plays an important role in the process and, in space, its action must be replaced by some other force. This work explores the possible usefulness of acoustic forces and, to a lesser extent, of an imposed liquid flow.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	1. We have documented in the earlier reports the unsatisfactory state of the art of numerical methods able to deal with free fluid surfaces in the presence of phase change. We have also described our own new method to deal with problems of this type. As with any new numerical method for complex problems, the issue of validation is of paramount importance. While of course agreement with experiment is essential, usually experimental data lack the detail which is desirable for a stringent validation. The most stringent test is a comparison with other independent codes implementing different numerical methods. For this purpose we have developed a new boundary-fitted method and code. A new feature of the fluid dynamic solver is second-order accuracy in space and time. Achieving this result has required the development of a new version of the projection method suitable for free surfaces. As for the temperature field, in order to have a sufficient resolution, the grid used for the energy equation is obtained by bisecting that used for the momentum equation. We are in the process of preparing two publications, one to describe the new algorithm and one to describe some results for vapor bubbles. 2. One of the most prevalent flow regimes in microgravity is slug flow, in which large Taylor bubbles alternate with liquid "slugs." It has long been known that Taylor bubbles exhibit significant stability features: what limits their length? Why do they lose axial symmetry in certain cases? What causes their coalescence? To answer these questions we have undertaken a study of several stability aspects of Taylor bubbles. Smoothness of the flow fields has proven to be essential to obtain good results. This has forced us to develop novel methods to deal with cells cut by the free surface. We have already spent quite a bit of effort to investigate the stability against non-axisymmetric perturbations and hope to complete the project in the near future. The problem is extremely complex as the accurate calculation of the eigenvalues requires excellent accuracy of the flow fields, which it is very difficult to achieve in free-surface problems of this type. Nevertheless, we have made substantial progress on this project and hope to complete it soon. 3. In addition to acoustic fields, which we have considered in several publications in the course of this project, electric fields may also be useful to control bubbles in liquids in microgravity. We have started a project to numerically examine this interaction in several cases of interest. In particular we have identified a new procedure to move bubbles in a liquid by coupling sound and electric fields oscillating at suitable frequencies.
Bibliography Type:	Description: (Last Updated: 10/29/2019)
Articles in Peer-reviewed Journals	Prosperetti A. "Bubbles." Physics of Fluids 16, 1852 (2004). https://doi.org/10.1063/1.1695308 , Apr-2004
Articles in Peer-reviewed Journals	Hao Y, Prosperetti A. "A numerical method for three-dimensional gas-liquid flow computations three-dimensional gas-liquid flow computations." J. Comput. Phys. vol. 196 pp. 126-144. https://doi.org/10.1016/j.jcp.2003.10.032 , May-2004